

FINA4370A Derivatives Trading: Analysis and Strategies

UNIT 15: THE CAPSTONE PROJECT: DEVELOPING A PROFESSIONAL TRADING PROPOSAL

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August 12, 2026

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1 Introduction: From Theory to Practice

Throughout this course, we have explored the mechanics, pricing, and strategic application of a wide range of derivative instruments. We have analyzed risk from multiple perspectives, from the "Greeks" of a single option to the systemic failures of major financial institutions. This final unit is the culmination of that journey.

The goal of this capstone unit is to synthesize this knowledge into a single, practical skill: the ability to construct a professional trading proposal. A well-structured proposal is what separates professional, disciplined trading from amateur speculation. It forces the creator to think through every aspect of a trade, from the initial idea to the final exit plan, in a rigorous and holistic manner.

2 The Core Components of a Trading Proposal

A professional trading proposal is a formal document designed to convince a portfolio manager or risk committee to allocate capital to a specific trade. It must be clear, concise, and comprehensive. The following components are essential.

2.1 Executive Summary

This is a one-paragraph summary at the very top. It should briefly state the trade thesis, the instrument used, the expected return, and the maximum loss. A reader should be able to understand the entire proposal at a high level just from this summary.

2.2 Trade Thesis and Rationale

This is the "story" behind the trade. It answers the question: *Why does this opportunity exist?* A strong thesis must be based on a clear analytical edge. Examples include:

- **Macroeconomic View:** "The market is underpricing the probability of a Federal Reserve interest rate cut, as seen in Fed Funds futures (Unit 12). We believe new inflation data will force the Fed's hand."
- **Relative Value:** "The implied volatility of stock XYZ is significantly higher than its historical realized volatility and the volatility of its peers. We believe it will revert to the mean after its earnings announcement."
- **Event-Driven:** "A potential merger between company A and B is likely to be approved. The current price of company A's options does not fully reflect the probability of this binary event."

2.3 Instrument Selection and Trade Structure

Here, you justify the specific derivative used to express your thesis. Why an option instead of a future? Why a spread instead of a single leg? This section must include the precise details of the proposed trade.

- **Underlying Asset:** e.g., S&P 500 Index, Crude Oil (WTI), Apple Inc. (AAPL) stock.
- **Instrument:** e.g., European Call Option, Futures Contract, Credit Default Swap.
- **Specifics:** Notional amount, strike price(s), expiration date, contract month.
- **Justification:** "We are buying a call option rather than the stock itself to create leverage and strictly define our maximum loss. We are using a strike price of \$110 because it offers the best trade-off between delta and premium cost."

2.4 Comprehensive Risk Analysis

This is the most critical section. It requires applying the holistic risk framework from Unit 13 and the case studies from Unit 14.

- **Market Risk:** What is the primary market exposure? What is the maximum potential loss? (e.g., "The max loss is limited to the \$50,000 premium paid"). What are the key Greeks (Delta, Vega, Theta) and how will they change? What is the stop-loss level?
- **Model Risk:** What are the assumptions behind this trade? (e.g., "This trade assumes volatility will decline. If a systemic shock causes volatility to spike, the trade will lose money even if the price moves in our favor").
- **Liquidity Risk:** How liquid is this instrument? What is the bid-ask spread? Could we get trapped in the position in a crisis (the LTCM lesson)?
- **Counterparty Risk:** Is this an OTC or exchange-traded product? If OTC, who is the counterparty and what is their credit quality? Is there a netting and collateral agreement in place?
- **Operational Risk:** How will the trade be executed, booked, and monitored to avoid errors (the Barings lesson)?

2.5 Profit/Loss Scenarios and Exit Strategy

This section quantifies the potential outcomes.

- **Payoff Diagram:** Include a visual payoff diagram for any option strategies.
- **Scenario Analysis:** A table showing the expected P&L under a base case, best case, and worst-case scenario.
- **Exit Plan:** Be explicit. The trade will be closed when:
 - (a) The profit target is reached (e.g., +100% return on premium).
 - (b) The stop-loss is breached (e.g., -50% of premium paid).
 - (c) The thesis is invalidated (e.g., the Fed explicitly rules out a rate cut).

(d) A certain amount of time has passed (e.g., Theta decay becomes too costly).

3 How Much? Sizing a Position Against a Risk Budget

Every trade proposal specifies an instrument, a strike and an expiration. Few specify **how many**, and the ones that do rarely say why. Yet size is the only decision that determines whether a bad outcome is an inconvenience or the end of the mandate.

The Sizing Identity

$$\text{Number of units} = \frac{\text{Risk budget for this trade}}{\text{Loss per unit in the defined bad case}} \quad (1)$$

Everything hard is in the denominator, and it is a **different quantity for different structures**:

- **Defined-risk structures** (spreads, condors, collars): the denominator is the contractual maximum loss. It is known exactly.
- **Long premium** (calls, puts, straddles): the denominator is the premium. Also known exactly.
- **Undefined-risk structures** (naked or ratio short options): there is no maximum, so the denominator must come from a **stress scenario** — not from VaR.

The numerator is a policy decision, not a calculation. A common institutional convention is 25–100 bp of fund value per idea, so that no single trade can cost more than a routine bad month.

Example 15.1: Sizing Three Structures Off One Budget

Scenario: A \$50 million fund allows **50 bp per trade**, giving a risk budget of **\$250,000**. Three candidate structures, each on a \$100 stock with 100 shares per contract:

Structure	Loss per contract	Basis	Size
Iron condor, \$5 wings, \$1.60 credit	\$340	contractual max	735 spreads
Long 3-month ATM call	\$543	premium paid	460 contracts
Short ATM straddle	\$2,720	1987-style stress	91 straddles

The first two denominators are contractual. The third is a judgement, and it is where sizing usually goes wrong. Note, e.g., the third position size is $250,000 / 2,720 = 91.9$ straddles. **Always round down.** A risk budget is a hard ceiling, not a target: 92 straddles would commit \$250,240 and breach the limit by \$240 before the first tick. The trader sells **91 straddles**.

The trap. Suppose the short straddle were sized off its **10-day 99% VaR** of \$45,804 on 100 straddles — \$4.58 per share — rather than off the stress scenario. The budget would then support **545 straddles** instead of 91: **a position six times larger**, on an instrument whose loss is unbounded.

VaR is a threshold, not a maximum (Unit 13). Sizing off it means sizing off the loss you expect to exceed one time in a hundred, which is the wrong end of the distribution for a position that can lose without limit. **Size undefined-risk structures off a stress scenario in which spot and volatility move together, never off VaR.**

Designing an Exit

This unit's template suggests a stop-loss. For a derivatives position that instruction needs unpacking, because **a stop-loss does different work depending on the payoff.**

Four Exits, and When Each Applies

1. **Profit target.** Close when the thesis has paid. For a short-premium trade this is usually a fraction of maximum profit — taking 50–75% of the credit early converts a long tail of small gains into a realised one and releases the risk budget.
2. **Stop-loss.** Close when the loss reaches a threshold. See the caveats below — this is the exit most often specified and least often workable.
3. **Thesis invalidation.** Close when the reason for the trade stops being true, **regardless of P&L.** If you bought a straddle because earnings were due and the company pre-announces, the trade is over whether it is up or down. This is the most professional of the four and the one students omit.
4. **Time stop.** Close by a date. Essential for any long-premium position, where theta guarantees the cost of waiting.

Three reasons a stop-loss behaves differently on a convex payoff.

1. **On a long option, a stop-loss does not limit loss.** The maximum loss is already the premium, contractually. A stop at –50% of premium is not risk control; it is a decision to **recycle capital** rather than hold to expiry. Say which you mean. A proposal that lists a stop-loss on a long call as its risk mitigation has not identified a risk.
2. **On a short-gamma position, the stop may not fill.** That is precisely where a stop is needed and precisely where it is least reliable: the position loses fastest during a gap, and a gap is the event in which the market prints past the stop without trading at it. **Barings and AIG both had exit plans.** Size as though the stop will not fill, and treat it as a bonus if it does.
3. **Stop on the underlying, not on the option's price.** An option can lose 40%

of its value on a volatility move with the underlying unchanged — stopping there exists a thesis that is still intact. Define the trigger on the variable the thesis is about: a level in the underlying, a term-structure spread, a date.

What complete risk, scenario and exit sections must contain

Combining this with Unit 13, a capstone proposal's risk, scenario and exit sections should state:

1. the **risk budget** for the trade, in currency and as a share of the notional fund;
2. the **loss per unit** in the defined bad case, and whether that figure is contractual or a stress estimate;
3. the resulting **size**, shown as a division;
4. a **VaR with all three parameters**, plus an Expected Shortfall or worst case;
5. at least one **stress scenario with spot and volatility shocked together**;
6. all four **exits** — and for the stop-loss, the variable it triggers on and an acknowledgement of gap risk.

Six aspects. It is the difference between “max loss \$50,000” and a position someone would actually authorise.

4 Evidencing Liquidity: Open Interest and Volume

Every proposal in this course names strikes and expiries. Naming them is not the same as showing you could trade them. A strike that exists on a screen and a strike that will fill 20 contracts at a price close to the one you quoted are different objects, and the difference has ended more strategies than a wrong directional view ever has.

Two numbers on the option chain answer the question, and they answer different parts of it.

Open Interest

Open interest is the number of contracts in that exact series — underlying, expiry, strike, call or put — currently **outstanding**: opened and not yet closed, exercised or expired.

It is not a count of trades. Whether a trade changes open interest depends on what *both* sides were doing:

Buyer	Seller	Effect on open interest
opening a new long	opening a new short	+1
closing a short	closing a long	−1
opening a new long	closing a long	unchanged
closing a short	opening a new short	unchanged

In two of the four cases a trade occurs and open interest does not move. This is

why volume and the change in open interest are not the same number, and why neither can be derived from the other.

Open Interest versus Volume — Different Questions

	Open interest	Volume
Answers	Is there a position here at all?	Does anyone trade it <i>today</i> ?
Measures	Stock (a level)	Flow (a rate)
Published	Once a day, after the close, for the <i>previous</i> session	Continuously through the session
Fails when	Large but dormant — an old position nobody trades any more	Spiky — one 5,000-lot print is not a liquid market

Report both. High open interest with near-zero volume means a position exists that nobody is currently willing to trade — which is precisely the situation in which you cannot get out. High volume with low open interest usually means day-trading flow that will not be there tomorrow. Either one alone can be made to look reassuring.

Three properties that limit what open interest can support

- 1. It is stale.** Open interest is compiled by the clearing house overnight and published the next morning, describing the position *as at the previous close*. Any figure you quote intraday is at least one session old. Say so, and give the date of the figure.
- 2. It is unsigned.** Open interest is a count. It does not record who initiated, who is long, or who is short — only that the contracts exist. Two identical open interest figures can represent opposite market structures. **Any claim that begins “the large open interest at the 105 strike means that...” has already assumed something the number does not contain.**
- 3. It counts only what cleared on an exchange.** A material share of retail option flow is internalised by wholesalers off-exchange. Exchange open interest is a large sample of the market, not a census of it.
Within those three limits open interest is exactly what element 3d asks for: evidence that the series you propose to trade is one that other people demonstrably hold. Outside them it supports nothing.

Example 15.2: Evidencing Liquidity on a Two-Leg Spread

The proposal. A bull call spread on XYZ, trading at \$100. Buy the 100 call, sell the 105 call, 45 days to expiry, **20 spreads**. Chain as at the close of 14 October:

Leg	Bid	Ask	Mid	Open int.	Volume
Long 100 call	4.10	4.25	4.175	3,140	486
Short 105 call	2.05	2.15	2.100	5,720	1,203

Step 1 — is the size supportable? Twenty spreads is 20 contracts per leg: **0.6%** of open interest on the long leg and **0.35%** on the short leg; **4.1%** and **1.7%** of the previous session's volume. Both comfortable. A working rule of thumb — and it is a rule of thumb, not a result — is that size beyond a few percent of a series' daily volume should expect to move the quote; if your proposal depends on the threshold, measure it rather than quoting mine.

Step 2 — price it where you would actually trade. You pay the ask on what you buy and receive the bid on what you sell:

$$\text{net debit} = 4.25 - 2.05 = \mathbf{2.20} \quad \text{versus mid-to-mid} = 4.175 - 2.100 = 2.075$$

The **entry slippage is 0.125 per spread**, or 6.0% of the mid price of the structure. Quoting the trade at 2.075 — which is what a mid-price screen shows — understates the cost before anything has gone wrong.

Step 3 — cost the whole round trip. You must get out as well as in. Measure everything against the frictionless mid-to-mid benchmark, so nothing is counted twice. At \$0.65 per contract, two legs, opened and closed:

Entry slippage	$0.125 \times 100 \times 20$	\$250
Exit slippage	$0.125 \times 100 \times 20$	\$250
Commissions	$80 \text{ contract-legs} \times \0.65	\$ 52
Total friction versus the mid-price benchmark		\$552

Step 4 — put it against the trade, not against the notional. On the mid-price benchmark the spread costs 2.075 and its maximum value is 5.00, so the benchmark maximum gain is $(5.00 - 2.075) \times 100 \times 20 = \$5,850$. Capital at risk is the debit you actually pay, $2.20 \times 100 \times 20 = \$4,400$. So

$$\frac{552}{5,850} = \mathbf{9.4\% \text{ of maximum gain}}, \quad \frac{552}{4,400} = \mathbf{12.5\% \text{ of capital at risk}}.$$

This is what element 3d is asking for, and it takes four lines. Note what it changes. A profit target of **30% of maximum gain, stated net**, requires gross P&L of $0.30 \times 5,850 + 552 = \$2,307$ — which is **39% of maximum gain**, not 30%. A proposal that prices at the mid never discovers that its stated target is nine points further away than it looks.

Example 15.3: The Leg That Fails the Test

Same underlying, same expiry. A proposal considers financing the structure by selling the 130 call instead:

Leg	Bid	Ask	Mid	Open int.	Volume
Short 130 call	0.05	0.35	0.20	12	0

The quote is the whole story. The bid–ask spread is 0.30 on a mid of 0.20: **150% of the mid price.** Selling 20 of these at the bid raises $0.05 \times 100 \times 20 = \100 , while buying them back at the ask would cost \$700. The exit costs seven times what the entry earned, before the market has moved at all.

And the two liquidity numbers agree. Open interest of 12 against a proposed 20 — **you would be the majority of the open interest in that series** — and zero volume in the last session. There is no market here; there is a quote obligation.

The correct treatment in a proposal is not to omit this leg. It is to show the numbers and say why you rejected it: *“the 130 strike was considered and rejected: bid–ask 0.05/0.35 is 150% of mid, open interest 12 against a required 20, no volume on 14 October. The \$100 of premium does not compensate for an exit that is not reliably available.”* That sentence is worth more marks than the trade it rejects.

What you may and may not claim from an option chain

May claim — these are facts about the chain:

- Open interest and volume at each leg, with the source and the date.
- The bid, the ask, the spread, and the spread as a percentage of mid.
- Your size as a percentage of open interest and of recent volume.
- Where open interest is concentrated relative to your strikes, *described as a concentration* — “open interest is heaviest at the 105 strike, 5% above spot and inside my target range”.

May not claim — the chain does not contain these:

- Who is long and who is short. Open interest is unsigned.
- That a concentration will act as support or resistance. That is a forecast resting on an assumption about dealer positioning which the chain cannot supply. (Unit 16, if read, sets out why.)
- That yesterday’s open interest describes today’s position.
- That exchange open interest is the whole market.

The asymmetry is the useful part. Liquidity evidence is strong grounds to **rule a strike out**, as in Example 15.3, and weak grounds to rule one in. Use it in the direction it works.

Where to get the numbers

Interactive Brokers, which this course already requires. Open interest arrives as tick types **27** (call) and **28** (put), requested via generic tick 101; the Greeks and implied volatility arrive as tick 13. Note the default limit of 100 simultaneous market data lines — request the specific series you need, not a whole chain.

Cboe publishes a delayed chain in JSON at

`cdn.cboe.com/api/global/delayed_quotes/options/{SYMBOL}.json` — indices take a leading underscore, e.g. `_SPX.json` — carrying strike, bid, ask, implied volatility, open interest, volume and first-order Greeks in one request, roughly fifteen minutes delayed.

Whichever you use, record the timestamp and put it in the proposal. A quote without a time is not evidence, and element 3a asks for the date and time of the quote in any case. Take one screenshot at the moment you price the trade; it costs nothing and settles every later question about what the market looked like.

5 A Professional Trading Proposal Template

Trading Proposal: [Trade Name]

Date: August 12, 2026

Trader: [Your Name]

1. Executive Summary

[One paragraph: We recommend a [long/short] position in [Underlying] using [Instrument] to express our view that [Thesis]. The target profit is X, and the maximum loss is Y.]

2. Trade Thesis / Rationale

[Detailed explanation of why the opportunity exists. Include supporting data, charts, or analysis. Reference relevant market conditions.]

3. Instrument Selection & Trade Structure

- **Underlying:**
- **Instrument:**
- **Notional / Size:**
- **Strike(s):**
- **Expiration:**
- **Justification:** *[Explain why this structure is optimal.]*

4. Comprehensive Risk Analysis

- **Market Risk (Max Loss / Stop-Loss):**
- **Key Greeks (Delta, Vega, etc.):**

- **Model Risk / Key Assumptions:**
- **Liquidity Risk:** bid/ask at each leg; open interest and recent volume at each leg; source and timestamp; estimated round-trip friction as a percentage of maximum gain.
- **Counterparty Risk (if OTC):**
- **Operational Risk:**

5. P&L Scenarios & Exit Strategy

- **Profit Target:**
- **Stop-Loss Trigger:**
- **Thesis Invalidation Condition:**

	Scenario	Expected P&L
• Scenario Table:	Best Case (Price = X)	
	Base Case (Price = Y)	
	Worst Case (Price = Z)	

6 End-of-Unit Exercises: Build Your Proposal

For this final unit, the exercises require you to use the template provided to create a detailed outline for a trading proposal based on the following scenarios. You do not need to use live market data; you can use the hypothetical data provided.

1. **Macro Policy Trade:** The market is pricing in a 50% chance of a 25 basis point interest rate hike by the European Central Bank (ECB) next month. You believe that recent weak economic data makes this highly unlikely, and the probability is closer to 0%. Outline a trading proposal to profit from this view using futures on a short-term interest rate benchmark (like EURIBOR).
2. **Equity Volatility Trade:** A major pharmaceutical company, "PharmaCo," is awaiting a key FDA ruling on its blockbuster drug in two months. The implied volatility on PharmaCo options is extremely high (e.g., 80%) as the market anticipates a massive price move. You believe the volatility is overpriced, but you do not have a view on the direction of the ruling. Outline a proposal for a trade that profits if the actual stock move is less than the market implies.
3. **Corporate Hedging Proposal:** You are the treasurer of a U.S.-based airline that has just ordered 10 new airplanes from Airbus, based in Europe. You have a massive payable of **€500 million** due in one year. The current EUR/USD spot rate is 1.0800. You are concerned the Euro will strengthen, increasing your USD costs, but you also want to benefit if the Euro weakens. Outline a proposal to the CFO recommending a specific derivative strategy to manage this currency risk. Compare your chosen strategy to at least one alternative.
4. **Sizing and Exits:** You manage a \$200 million fund with a 40 bp per-trade risk

budget. You propose selling a put spread on an index: sell the 95-strike, buy the 90-strike, \$5 wide, for a \$1.75 credit per share (100 shares per contract).

- (a) Compute the risk budget and the position size.
- (b) You instead propose selling naked 95-strike puts for a \$2.60 credit. Explain how the sizing calculation changes, and what figure you would put in the denominator.
- (c) Specify all four exits for the put spread.
- (d) Your colleague argues the naked put is better because it collects a larger credit. Respond in terms of risk-adjusted size.